MCQs FILE OF BIO401

FINAL TERM

CREATED BY JIYYA RANA

1) Name two types of statistics...........

Discriptive

inferential

2) Name two types of Discriptive statistics...........

Graphical measure

Numerical measure

3) Name two types of Numerical measure......

Frequency distribution

Summary measure

4) Name two types of summary measure.......

Measure of central tendancy

Measure of dispersion

5) Name three types of Measure of tendency......

Mean

Median

Mode

6) The extent to which the observation in the data set are spread away from the central value is called..........

Edit with WPS Office

Dispersion

7) ....are the method that convey information regarding the amount of variability present in the set of data.

Measure of dispersion

8) Variation, Spread, and Scatter are the term used .... with dispersion.

Synonymusly

9) PPO is the abrivation of.........

Polyphenolic Oxidase

10) YPC is the abrivation of...........

Yellow Pigment Content

11) How many types of measure of dispersion?

Two

12) Name two types of measure of dispersion.....

Relative measure of dispersion

Absolute measure of dispersion

13) Measures which are expressed in term of same unit as original data are termed as............

Absolute measure of dispersion

14) Measures which are expressed in term of ratios and %age are termed as.........

Relative measure of dispersion

IMPORTANT POINT

- Absolute measure of dispersion can converted into relative measure of disperssion but relative measure of dispersion cannot be converted into abosolute measure of dispersion

Edit with WPS Office

15) Which of the following is not the type of absolute measure of dispersion......

a) Range

b) Quartile deviation

c) Mean

d) Mean absolute deviation

e) Standard deviation and variance

16) The difference b/w minimum and maximum value in the data is called........

Range

17) .... is the half difference b/w upper quartile and lower quartile?

Quartile deviation (Q.D)

18) Formula for measuring the lower quartile?

$Q_1 = 1(n+1)/4$ value

19) Formula for measuring the upper quartile?

$Q_3 = 3(n+1)/4$ value

20) Formula for quartile deviation?

Q.D = Inter quartile range /2 $= Q_3 - Q_1 /2$

21) IQR is the abrivation of........

Inter quartile range

22) .... of the absolute deviation measured either from mean or median.

Arithmetic mean

23) .... is define as the mean of the squared deviation of the observation from thier mean.

Edit with WPS Office

Variance

24) The square root of the variance is called..........

Standard daviation

25) The variance is independent of the ....

Origin

26) Comparing dispersion in two data sets on the basis of standard deviation may lead to ....

fallacious results.

27) C.V is the abrivation of ....

Coefficient of variation

28) ....is define as the deperture of symmetry.

Skewness

29) What is the symbol of skewness?

S.K

30) What is the formula of skewness?

S.K = Mean - mode / S.D

31) The useful visual device for communicating the information contained in a data set is called.........

Boxplot

32) Box and whisker plot drawn for .... variable

Quantitative variables

33) Some activity with observable outcomes is called........

Experiment

Edit with WPS Office

34) An experiments which produced different outcomes in multiple repititions of the experiment, performed under similar condition is called...........

Randome experiment

35) Rolling of a dice is the example of .... experiment

Random

36) How many types of events?

two

37) Name two types of events...

Simple and composite event

38) .... event consist of only one outcome..

Simple

39) If event consist of more than one outcomes it is called........ event.

composite

40) Name varoius other types of events.......

Mutually exclusive event

Collectively exhaustive event

Equally likely event

Indepedent and dependent event

41) How many types of probability?

Two

Edit with WPS Office

42) Name two types of probabilty

Objective
Subjective

43) Name further two types of objective probability....

Classical
relative frequency

44) In .... year L.J Savage give the concept of probability.

1950

45) Two or more events are said to be .... event if the event do not have any outcomes in common?

Mutually exclusive event

46) Two or more events are said to be ....event if one event is as likely to occur as the other?

Equally likely event

47) When the occurance of event A has an influence on the occurance of an event B then this event is called

Dependent event

48) When the occurance of event A has no effect on event B then this event is called.......

independent event

IMPORTANT POINT

- Complementary event are always mutually exclusive event but mutually exclusive event are not complementary event

49) In .... year the axiomatic approach to probabilty was

Edit with WPS Office

formalized by the Russian Mathmatician A.N Kolmogrove.

1933

50) Three axiomatic properties of probability.......

- Axiom of Non — Negativity
- Axiom of Exhaustiveness
- Axiom of Additive - ness

51) If the some of the mutually exclusive event is equal to 1 then this called.........

Axiom of Exhautiveness

52. Formula that use to find probability that this person will be 18 year old or younger ....?

Ans: p (E)=number of early subject /total number of subject

53. The probability we are seeking may be written in symbolic nation as.........

Ans: p (E intersection A)

54. The symbol intersection show that............

Joint occurrence

55. The formula of joint probability ....

Ans: n (E intersectin A)/n (s)

56. Formula of conditional probability..................is

P (A/B) =P (A intersection B)/P(B)

57. Which condition show a conditional probability...........

A vertical line .between A and B

58. When the occurance of an event A has <u>no effect</u> on the occurance of an event B is called..................?

Edit with WPS Office

Independent Event

59. When the occurance of event A has an <u>influence</u> on the occurance of an event B. that is called ……………?

Dependent event

60. The way to find the probability of two event occurring at the same time is called……………..?

Multipicative rule

61. Formula of multiplicative rule…………is?

P (A and E) = P (A intersection B)

62. Multipicative rule is also called …..?

AND rule

63.For any **two independent event** A and E,the probability for the **joint occurance** of the both A and E is the product of their individual probabilities…..is a

.Specific Multipicative rule

64. Each parents has ……………… chance of making an (a) gamete

½

65. If the event do not have an outcomes in commen .They are the event **that cannot occure together.** **….is?**

Mutually exclusive

66. The event that occure together is……

Not-mutually event

67. Following **TWO** rules use to calculate the probability an event……….?

- .Rule of complementary event
- Bayes rule

68. A……………is used to determine the presence or absence of a

Edit with WPS Office

disease.

Diagnostic test

69. A…………. identifies a symptotic individual who may have disease.

Screening test

70. Blood pressure for hypertension is a……………..?

Screening test

71. PSA test for prostate cancer……….is

Screening test

72. Fasting blood suger for diabtes is a……?

Screening test

73. Blood glucose level……….

70-100mg/dL

74.Cholesterole level………is

Less than 100mg/dL

75. Systolic blood pressure is…….?

120

76. Diastolic blood pressure is…?

80

77. When test indicate a positive status……….

Having a disease

78. When test show a negative status it indicate ……

Not having disease

79. Sensitivity and specificity of a diagnosis test depend on the…………..?

Edit with WPS Office

Quality of test

80. ROC curves is reported by the …………….?

Goldestien and Mushline

81. The lower the <u>T4 values</u>, show the………

Patients are to be Hypothyroid

82. ROC analysis a part of the field called…………..?

Signal detection theory

83. The portion that develops disease is a ……?

Incidence disease

84. If you report the incidence rate of say the heart disease study as ………per………person years.

2.5 , 1000

85. Prevalence is measure for following disease………..?

Chronic disease such as diabetes or Osteoarthritis

86. …………….. is measure of frequency of occurrence of death.

Mortality

87. IS the infant mortality rate is a raio?

Yes

88. An experiment or a trial , whose outcome can be classified as either a success or a failure is called……………..?

Bernoulli trial

89. Jacob Bernoulli is a…………….?

Swiss Mathematician

90. Jacob Bernoulli was born………and death in...... year?

1705-1965

Edit with WPS Office

91. X=1

When the outcome is success

92. When the outcome is failure……………?

X=0

93. A sequence of Bernouli trials forms a ------------?

Bernouli process

94.Each Bernouli trials results in one of the ---------- possible outcomes?

Two

95. The name of two possible outcomes are?

- Failure
- Success

96. The probability of success denoted by --------?

P

97.The probability of success remains -------- from trial to trial?

Constant

98.The trials are ----------?

Independent

99.A ------------------ results from Bernouli process?

Bionomial probability Distribution

100.A bionomial probability of distribution has ---------- number of trials?

Edit with WPS Office

Fixed

101. ---------- will denote the probabilities of Sucess and Failures respectively?

P and q

102. P is ---------?

Probability of success

103. q is ----------?

Probability of failure

104. n is a fixed number of -------?

Trials

105. x is a ---------?

Specific number of success in n trials

106. P(x) is equal to --------?

Probability of getting exactly x success among the n trials

107. $\boxed{p^x(1-p)^{n-x}}$ = is a ---------?

Probability of x success among n

108. How many methods for finding Probability?

Three

109. In a --------, we finding the probabilities corresponding to the random variable x ?

Bionomial Distribution

110. Methods of finding probability?

Edit with WPS Office

- Using the bionomial probability formula
- Using the Table of probabilities
- Using technology

111. Suppose we have n = 5 and *p* = 0.60 The probability that bionomial trials yeild 4 success will be given as :

Ppppq = (0.60)4 x 0.40 = 0.05184

112. One has to be very certain that x and p both refers to the same cetagory being called a ----------?

Success

113. When sampling is conducted without replacement, then consider events to be -------- if n<0.05 n:

Independent

114. The -------- suggests that values are unusual if they are outside of these limits.

Range rule of Thumb

115. $\mu + 2\sigma$ is a ----------- value.

Maximum value

116. $\mu - 2\sigma$ is a ---------- value.

Minimum Value

117. For p = 0.5 and large or small n the bionomial distribution will be:

Symmetric

118. For large p and small n the bionomial distribution will

Edit with WPS Office

be ------?

Left skewed

119. For small p and small n the bionomial distribution will be -------?

Right skewed

120. For small p but large n the bionomial distribution approaches ------?

Symmetry

121. By using MS excel we will use a function

BINOMDIST

122. In interest, number of events occurring ?

2

123. Number of Deaths claims received per day by a -------- company?

Insurance

124. Number of Alpha particles emitted from a ----------- during a given period of time?

Radioactive source

125. The discrete distribution used for such instances; number of families with child, About deaths claims and alpha particles emitted from radioactive source is called --------?

Poisson probability distribution

126. Who invented the poisson distribution?

Edit with WPS Office

French mathematician Simeon Denis Poisson

127. The probability that event occurs in a given unit of time is the ----- for all units?

Same

128. The events that occur in one unit of time is -------- of the number of events in other units.

Independent

129. The mean or expected rate Is ----?

λ. (lamda)

130. λ. Is -----?

Average number of events per interval

131. The base of natural logarithms is --------?

Eulers number

132. $p(x) = \frac{\lambda^x}{x!} e^{-\lambda}, \qquad x = 0, 1, 2, \ldots$ the following equation is called ------?

Poisson Random variable

133. The poisson process generates a -------?

Poisson distribution

134. How many traits in poisson process ?

Three

135. In poisson process -------- are discrete??

Outcomes

136. The probability of two or more successes over a

Edit with WPS Office

sufficiently small interval is essentially --------?

Zero

137. In the study drug induced of anaphyl axis among patients taking -------- as a part of their anesthesia?

Recuronium bromide

138. $P(X = 3) = \frac{e^{-12}12^3}{3!} = .00177$ it is the -------- of patients receiving recuronium?

Probability

139. poisson distribution is often used as an ------ for bionomial probabilities ?

Approximation

140. Poisson distribution is important because it is often used for discribing the behaviour of ----------?

Rare events

141. A --------- is a collection of all elements of interest?

Population

142. A -------- is a subset of population?

Sample

143. kinds of sample ??

Representive or non-representive

144. --------- samples are the samples selected using the principles of probability?

Probabilistic

Edit with WPS Office

145. The researcher can determine the probability of any given sampling error and make ------------- about population characteristics.

Statistical interferences

146. --------- sample may be resonably representative?

Quota sample

147. Sample for each stratum or subgroup of a population is called.......

Startified samples

148. Combinations of random , systematic ,startified and cluster sampling is called........

Multistage sampling

149. If probability involves at each stage then which sample distribution can be obtained?

Sample statistics

150. The -------- CMAs of Regina and Saskatoon are strata?

Two

151. Lists of elements that sample selected from is called -----?

Frame

152. Characteristics of a population is called ?

Parameter

153. Numerical characteristics of a population is called?

Statics

Edit with WPS Office

154. Probability of distribution of a static is called -------?

Sample distribution

155. In -------- each sample possible of size *n* has the same probability of being selected.

Simple random sample

156. It may be difficult to obtain a -------?

Frame

157. The --------- is the frequency of occurence of a characteristic divided by total number of elements.

Proportion

158. ---------- may also be termed as population values ?

Parameters

159. A specific value of point estimator is reffered as ---------- of a parameter.

Point estimator

160. Difference between the point estimate and the value of parameter is called ?

Sampling Error

161. Culter poll has a sampling error of only ------ % point for the liberals.

0.2%

162. A ----- is the probability diatribution for all possible values of sample static.

Sample distribution

Edit with WPS Office

163. In sampling distribution each sample contain ------- elements.

Different

164) ________ is a statistical technique used to determine the degree to which two variables are related.

Correlation

165) Name of two quantitative variables

One variable is called independent (X) and

The second is called dependent (Y)

166) measures the nature and strength between two variables of

the ______ type

Quantitative.

167) The sign of .... denotes the nature of association

r

168) The value of ....denotes the strength of association.

r

169) Statistic showing the degree of relation between two variables is also called

Pearson's correlation or product moment correlation

coefficient.

170) If the sign is +ve this means the relation is ....

Edit with WPS Office

direct.

171) While if the sign is ....this means an inverse or indirect relationship.

-ve

172) Regression tells us how to draw the straight line described by the ....

correlation.

173) .... describes the strength of a linear relationship between two variables.

Correlation

174) If r = ... this means no association or correlation between the two variables.

Zero

175) If 0 < r < 0.25 = ...correlation.

weak

176) If 0.25 ≤ r < 0.75 = ...correlation.

intermediate

177) If 0.75 ≤ r < 1 = ...correlation.

strong

178) If r = l = ... correlation.

perfect

179) Spearman Rank Correlation Coefficient (rs)

It is a non-parametric measure of correlation.

Edit with WPS Office

180) The names of four windows.

Data editor

Output viewer

Syntax editor

Script window

181) SPSS stands for............

Statistical Product and Service Solutions.

182) .... is Spreadsheet-like system for defining, entering, editing, and displaying data. Extension of the saved file will be sav.

Data Editor

183) Displays output and errors. Extension of the saved file will be spv.

Output Viewer

184) The.......shows that more people who receives wireless service tends to own PDA compared to people who doesnt receive wireless service.

graph

185) ....Text editor for syntax composition. Extension of the saved file will be sps.

Syntax Editor

186) ....Provides the opportunity to write full-blown programs, in a BASIC-like language. Text editor for syntax composition. Extension of the saved file will be sbs.

Edit with WPS Office

Script Window

187) There are two sheets in the window:

1. Data view 2. Variable view

188) The basic analysis of SPSS that will be introduced in this class.

— Frequencies

— Descriptives

— Linear regression analysis

Edit with WPS Office

Stat 501 MCQs

1) ___ is a subset of the population (ans)

Sample

2) The collection of all the element of interest is ___

Population

3) Large sample would offer the value of population

4) ___ obtain a true value of population

Whole sample

5) A sampling ___ in which each plot member has an equal chance of being selected is called

Random Sampling

6) ___ is a list of element that sample selected from

Frame

Edit with WPS Office

7) In which case the element has the same chance of being selected when

$n = 2$

8) If A, B, C, D then how many combinations will be possible.

6 combination;

9) Formula used to find combination

$$C = \frac{N!}{n!\ ?}$$

10) ______ preferred for sampling

careful design.

11) ______ is used for standard deviation

σ

12) μ is ______

mean

(a) ___ characteristics of a population

Parameter

(b) Parameter also termed as population value

(c) Difference between the point estimate and the value of parameter is ___ ?

standard error

(d) probability distribution for all the possible values of the sample statistics is

Sampling distribution

(e) How Sampling method may allow the researcher to determine the Sampling ___ ?

Edit with WPS Office

when sampling is selected

ii) The mean of distribution of $\bar{x}$ is

μ

iii) ____ referred to as

point estimator

iv) SDP stand for

sampling distribution sample

Edit with WPS Office

THE END

REMEMBER ME IN YOUR PRAYER

REGARDS = JIYYA RANA

Edit with WPS Office

Edit with WPS Office